

*AI Regtech
Global Standard Setter
Top Regulators Validation*



Aereve makes **UK-FCA**

AI Live Test Cohort 2 (Mature Product Only)

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• CASE STUDY • AML / KYC • GENERATIVE AI

Enhancing AML/KYC with Generative AI in a Regulated Banking Environment

Aereve partnered with Citibank to deploy a fine-tuned GenAI model for AML screening — validated inside the HKMA GenAI Sandbox and powered by Cyberport's AISC H800 GPU infrastructure. Aereve is the **only Cyberport tech firm** admitted into Cohort 1.

81.2%

RELEVANCE ACCURACY

92.4%

SENTIMENT ACCURACY

69%

ROUGE-L SCORE

~80%

TIME & RESOURCE REDUCTION

500+

REAL-WORLD AML CASES

● THE CHALLENGE

Citibank faced a surge in **false positives** driven by Chinese character name variations (e.g., 峰 / 峯), making manual AML screening time-consuming, error-prone, and resource-intensive at scale. Sharing bank data securely required significant infrastructure investment, while producing quality training data demanded extensive human labelling effort.

False Positives

Manual Review

Data Security

Chinese Name Variants

● THE SOLUTION

Aereve fine-tuned a **Llama-based GenAI model** on 500 real-world AML cases using Cyberport AISC's H800 GPU cluster, delivering a domain-adapted model that outperforms both baseline Llama and DeepSeek-V3 on all three AML/KYC tasks. The on-premise deployment ensured full compliance with Citibank's information security requirements throughout the sandbox process.

Fine-Tuning (LoRA)

On-Premise Deploy

H800 GPU Cluster

Llama 3.2-3B

● THREE CORE AML / KYC TASKS AUTOMATED BY GENAI

TASK 01

Relevance Classification

Determines whether matched information refers to the same entity — reducing false positive workload at the first triage stage. Fine-tuned model achieves **81.2%** accuracy, surpassing DeepSeek-V3 (75.7%) and baseline Llama (66.5%).

TASK 02

Sentiment Analysis

Classifies news and article sentiment as Positive / Neutral / Negative to support risk-level assessment. Achieves **92.4%** weighted F1-score, with 94.6% precision on positive and 93.2% on negative classifications.

TASK 03

Text Summarisation

Extracts key AML/KYC information and risk indicators from unstructured data. Fine-tuned model achieves **ROUGE-L 69%** — a 21.7% improvement over baseline — enabling faster, more consistent analyst reviews.

● MODEL PERFORMANCE COMPARISON

RELEVANCE CLASSIFICATION ACCURACY

Baseline Llama	66.5%
DeepSeek-V3	75.7%
Fine-tuned (Ours)	81.2%

SENTIMENT ANALYSIS F1-SCORE

Baseline Llama	~70%
Fine-tuned (Ours)	89.7%

Domain fine-tuning on 500 real AML cases with LoRA on Cyberport H800 GPUs yields consistent gains over both general-purpose baselines across all three tasks.



If we start again, are we going to do it given all the hurdles? YES! Invaluable learning experience. The 80/20 Rule: if we can potentially achieve an 80% reduction of time and resource, why not? The fine-tuned model demonstrates substantial improvement over the baseline — particularly for classification and sentiment analysis tasks.

Edgar Ma

Cluster AML Head of Greater China and Korea, Citibank

● HKMA GENAI SANDBOX CONTEXT

The HKMA and Cyberport jointly launched the **GenAI Sandbox** to foster responsible AI innovation in banking — covering risk management, anti-fraud, and customer experience. Cohort 1 comprised **15 use cases** from 10 banks and 4 technology partners, selected from over 40 submissions. Aereve is the **only Cyberport tech firm** admitted alongside major banking institutions.

15 Use Cases

10 Banks

4 Tech Partners

40+ Submissions

● SOURCE DOCUMENTS

**Citibank & Aereve: HKMA GenAI Sandbox Success**

Validated in HKMA Sandbox | Supported by Cyberport AISC

Citibank: The Challenge & Why Sandbox

High Volume of False Positives: The increasing use of Chinese names in our processes, particularly due to character variations (e.g., 峰 / 峯), exacerbates the false positive problem

HKMA IPMN Scheme: Partnered with HK Baptist University and Aereve as dedicated mentor, nurturing FinTech talent through practical industry experience

Manual Review Inefficiency: Manual reviews of large volumes of unstructured data are time-consuming, error-prone, and resource-intensive

Cyberport AISC: Leveraged H800 GPU resources for model fine-tuning on 500 real-world AML cases

Hurdles Encountered

- Information Security: Sharing bank data is difficult; establishing secure server required significant investment
- Data Labelling: Extensive human intervention needed for quality training data

Aereve Report: https://aereve.com/assets/Aereve_Citi_HKMA_Sandbox.pdf

HKMA Report: <https://brdr.hkma.gov.hk/eng/doc-ldg/docld/getPdf/20251031-6-EN/20251031-6-EN.pdf>

**HKMA Gen A.I. Sandbox Final Report**

Enhancing On-Premise AML/KYC System with GenAI:
Improving Accuracy, Efficiency, and Compliance

by Citi & Aereve 2025-08-18

3. Solution Overview

3.1 Data Collection and Preparation

- Data Source:** The dataset comprised the input customer profile and google search results of their web, social media, third-party entities, etc. to match data, and the targeted data has approximately 1,000 GB in size. The information gathered for each part of dataset was as follows:
 - Input profile: entity name, bank, country, gender, date of birth, occupation, etc.
 - Search results: google link, snippet, and link content (extracted search results as shown in Fig. 4 below).



Fig. 3 Screenshot of data collection interface for the customer profile input



Fig. 4 Screenshot of data collection interface for the customer profile input

The questions of each part of data include three important AML/KYC tasks including:

- Relevance Classification:** determine whether the matched information is referring to the entity based on the input data.
- Sentiment Analysis:** determine the sentiment of the matched information.
- Text Summarisation:** determine the key information of the matched entity (e.g., assets, relationships, etc.).